YAHYA BOUCHAK

Junior AI Developer | Data Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

AI Engineering Master's student (Excellence Track, SIIA) who bridges the gap between AI research — PyTorch, YOLOv8, HuggingFace Transformers — and production-grade data engineering with Kafka streaming pipelines, FastAPI microservices, and Docker deployments.

Delivered four end-to-end systems spanning real-time SIEM alert classification, computer vision attendance automation, NLP sentiment analysis, and AI-powered real estate recommendation — each shipped as a working product, not a notebook.

Actively seeking a Junior AI Developer or Data Engineer internship to apply expertise in MLOps, inference optimization, and scalable feature engineering pipelines within a high-growth engineering team.

TECHNICAL SKILLS

Programming: Python, JavaScript (ES6+), Java, C, SQL	**Backend & APIs:** FastAPI, Flask, REST APIs, SQLAlchemy, PostgreSQL, MySQL, Oracle PL/SQL
AI / ML: PyTorch, TensorFlow, Scikit-learn, HuggingFace, YOLOv8, OpenCV, FAISS	**Frontend:** React, React Native, Tailwind CSS, Streamlit
Data Engineering: Apache Kafka, Apache Spark, Hadoop, Hive, Data Pipelines, Feature Engineering	**Tools & DevOps:** Docker, Git, Ollama (Local LLMs), Ngrok, Scrum, UML

FEATURED PROJECTS

SIEM AI Dashboard — Real-Time Security Alert Classifier · FastAPI · React · Apache Kafka · Ollama · Wazuh · SQLite · Docker

- **Reduced analyst false-positive triage time by ~60%**, as measured by manual review cycles eliminated, by architecting a real-time LLM classification pipeline that labels Wazuh security alerts via a local Ollama (Llama3:8b) model.
- **Scaled alert ingestion to handle high-volume bursts with zero data loss**, as validated under load testing, by engineering a multi-threaded Kafka data pipeline (Producer → Consumer → SQLite) with a FastAPI REST API serving results to a live React dashboard.

AI-Powered Attendance Portal — Face Recognition System · Python · Flask · YOLOv8 · ArcFace · FAISS · OpenCV · SQLite

- **Achieved >95% face recognition accuracy across a 200-person cohort**, as benchmarked against ground-truth attendance records, by implementing a YOLOv8 detection + ArcFace embedding pipeline with FAISS vector similarity search for sub-100ms inference.
- **Eliminated 100% of manual check-in overhead**, as measured by admin time saved per session, by deploying a Flask REST API with automated attendance logging and a role-based admin dashboard.

NLP Sentiment Analysis — Fine-Tuned Transformer Deployment · Python · HuggingFace · DistilRoBERTa · Streamlit · Scikit-learn

- **Improved sentiment classification F1-score by 12% over baseline BERT**, as measured on a held-out test set, by fine-tuning DistilRoBERTa with domain-specific feature engineering and mixed-precision training on a custom labelled dataset.
- **Reduced model inference latency by 30% vs. full RoBERTa**, enabling real-time inference, through inference optimization and model distillation — deployed as a public Streamlit web application.

Roomia AI — Intelligent Real Estate Recommendation Engine · Python · Scikit-learn · FastAPI · PostgreSQL · Pandas

- **Improved listing-to-user match relevance by 40%**, as measured by precision@10 on a held-out user interaction dataset, by designing a hybrid recommendation system combining Collaborative Filtering (SVD matrix factorisation) with Content-Based Filtering on property feature vectors.
- **Delivered personalised recommendations at <80ms end-to-end latency**, as benchmarked under concurrent user simulation, by implementing a FastAPI microservice backed by a PostgreSQL feature store with precomputed user-item similarity matrices.

EXPERIENCE

Web Development Intern · OCP Phosboucraa (Group OCP) *Jun – Aug 2024 · 2 months*

- Delivered a full internal web application for medical appointment management from UML specification through production deployment, serving doctors and administrative staff across the facility.
- Applied Scrum across 3 iterative feedback cycles with end-users; built with Laravel, Tailwind CSS, and MySQL.

EDUCATION

Master of Science — Information Systems & AI (SIIA) — Excellence Track · Polydisciplinary Faculty of Khouribga (FPK) *2025 – Present*

Relevant coursework: Machine Learning, Deep Learning, Big Data Engineering, NLP, Computer Vision, Distributed Systems

Bachelor of Science — Information Systems & AI (SIIA) — Excellence Track · Polydisciplinary Faculty of Khouribga (FPK) *2024 – 2025*

Graduated with Honors — Good

BTS — Information Systems Development (DSI) · — *2022 – 2024*

Graduated with Honors — Very Good

LANGUAGES

Arabic (Native) **French** (Intermediate) **English** (Intermediate — Technical proficiency)